

Mako / Optiver / Jane Street – Brutal+ Assessment

Batch 1 (Q1–Q10) Time: 10–12 min

Q1. Start at 0. Each step: +1 w.p. $\frac{1}{2}$, reset to 0 w.p. $\frac{1}{2}$. Stop when reaching 3. Expected steps?

- A. 8
- B. 10
- C. 12
- D. 14
- E. 16

Q2. Given $x + \frac{1}{x} = 3$. Compute $x^5 + \frac{1}{x^5}$.

- A. 123
- B. 153
- C. 183
- D. 213
- E. 243

Q3. $a_{n+1} = (a_n + 4)/3$, $a_1 = 1$. Limit?

- A. 1
- B. 2
- C. 3
- D. 4
- E. Diverges

Q4. Lose 20%, gain 25% repeatedly. Long-term capital?

- A. strictly increases
- B. strictly decreases
- C. stays constant
- D. oscillates
- E. diverges

Q5. Solve $\frac{1}{x-2} + \frac{1}{x+2} = \frac{1}{2}$.

- A. $\pm 2\sqrt{2}$
- B. $\pm \sqrt{6}$
- C. ± 4
- D. ± 2
- E. No real solution

Q6. If $x+y+z=0$, evaluate $x^3+y^3+z^3-3xyz$.

- A. 0
- B. xyz
- C. $3xyz$
- D. $-3xyz$
- E. Cannot determine

Q7. +100%, -50%, +100%, -50%. Net multiplier?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 0.5

Q8. Approximate $(1.01)^{200}$.

- A. 5

- B. 6
- C. 7
- D. 8
- E. 9

Q9. $a(n+1)=a_n^2-2$, $a_1=2$. Find a_4 .

- A. 2
- B. 0
- C. -2
- D. 4
- E. 14

Q10. 2% loss per transfer over 20 transfers. Multiplier?

- A. 0.67
- B. 0.82
- C. 0.60
- D. 0.45
- E. 0.30

Brutal+ Assessment — Batch 2 (Q11–Q20)

Q11. A biased coin has $P(H)=p$. You flip until first head. Expected flips = 4. Find p .

- A. $1/2$
- B. $1/3$
- C. $1/4$
- D. $2/3$
- E. $3/4$

Q12. You draw cards until first Ace from a shuffled deck. Expected draws?

- A. 10
- B. 13
- C. 17
- D. 20
- E. 26

Q13. A game: roll die. If 1–5 → gain \$1 and repeat. If 6 → gain \$10 and stop. EV?

- A. 5
- B. 6
- C. 7
- D. 8
- E. 9

Q14. Two coins: fair and double-headed. Pick one at random. Flip once → Head. $P(\text{double-headed})$?

- A. $1/2$
- B. $2/3$
- C. $3/4$
- D. $1/3$
- E. $1/4$

Q15. Expected number of tosses to get first tail if $P(T)=0.25$?

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

Q16. You roll 2 dice. Given sum=7, probability one die is 2?

- A. $1/6$
- B. $1/5$
- C. $1/3$
- D. $2/5$
- E. $1/2$

Q17. A box has 3 red, 3 blue. Draw 2 without replacement. $P(\text{same color})$?

- A. $1/2$
- B. $2/5$
- C. $3/5$
- D. $4/5$
- E. $1/3$

Q18. Expected maximum of 2 uniform[0,1] variables?

- A. $1/2$
- B. $2/3$

- C. $3/4$
- D. $4/5$
- E. $5/6$

Q19. You flip until you see HT. Expected flips?

- A. 4
- B. 5
- C. 6
- D. 7
- E. 8

Q20. A test: 95% accurate, disease prevalence 2%. $P(\text{disease} \mid \text{positive})$?

- A. ~25%
- B. ~30%
- C. ~40%
- D. ~50%
- E. ~70%

Brutal+ Assessment — Batch 3 (Q21–Q30)

Q21. You play a game: with prob $1/3$ you gain $+3$ and continue; with prob $2/3$ you lose -2 and stop. What is the expected value?

- A. -1
- B. 0
- C. 1
- D. 2
- E. 3

Q22. Flip a fair coin until you see HTH. Expected flips?

- A. 6
- B. 7
- C. 8
- D. 9
- E. 10

Q23. A box has 2 red, 2 blue. Draw until first red. Expected draws?

- A. 1.5
- B. 2
- C. 2.5
- D. 3
- E. 3.5

Q24. You repeatedly roll a die until you get a number ≥ 5 . Expected rolls?

- A. 2
- B. 2.5
- C. 3
- D. 3.5
- E. 4

Q25. Two independent geometric variables with $p=0.5$. Expected minimum?

- A. 1
- B. $4/3$
- C. $3/2$
- D. 2
- E. $5/2$

Q26. You choose a random point in $[0,1]$. Expected distance to nearest endpoint?

- A. $1/4$
- B. $1/3$
- C. $1/2$
- D. $2/3$
- E. $3/4$

Q27. A Markov chain: stay in same state w.p. 0.7 , switch w.p. 0.3 . Long-term fraction in each state?

- A. depends on start
- B. $0.7/0.3$
- C. $0.5/0.5$
- D. $0.3/0.7$
- E. none

Q28. Expected number of coin tosses to get at least one H and one T?

- A. 2
- B. 3

- C. 4
- D. 5
- E. 6

Q29. Roll a die repeatedly. Expected value of the maximum after 2 rolls?

- A. 3.5
- B. 4
- C. 4.5
- D. 5
- E. 5.5

Q30. A test: 90% sensitivity, 90% specificity, prevalence 10%. $P(\text{disease} \mid \text{positive})$?

- A. 50%
- B. 60%
- C. 70%
- D. 80%
- E. 90%

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Batch 4: Q31-Q40 | Pattern Recognition + Invariants

Instructions: No calculator. Choose exactly one answer per question.

Q31. A sequence is generated by alternating two operations: multiply by 3, then add 1. Starting at 2, the visible sequence is 2, 6, 7, 21, 22, 66, ?. What comes next?

- A. 67
- B. 88
- C. 99
- D. 132
- E. 198

Q32. The sequence below is formed by a hidden rule involving the previous two terms and parity: 1, 2, 4, 7, 13, 24, ?. What is the next term?

- A. 37
- B. 41
- C. 44
- D. 46
- E. 48

Q33. For each term after the first, multiply the previous term by 2 and then subtract an increasing odd number: 3, 5, 7, 9, ... Starting at 10, the sequence is 10, 17, 29, 49, ?. What comes next?

- A. 75
- B. 79
- C. 81
- D. 83
- E. 89

Q34. Consider the sequence: 4, 9, 19, 39, 79, ?. The obvious doubling pattern has a hidden off-by-one trap. What is the next term?

- A. 149
- B. 155
- C. 159
- D. 161
- E. 163

Q35. Letters are generated by cumulative jumps: +2, +4, +8, +16, wrapping around the alphabet if needed. Starting from B: B, D, H, P, ?. What comes next?

- A. B
- B. E

- C. F
- D. G
- E. H

Q36. The sequence is formed by interleaving two subsequences: one arithmetic and one geometric. 3, 2, 7, 6, 11, 18, 15, ?. What is the missing next term?

- A. 24
- B. 36
- C. 42
- D. 54
- E. 72

Q37. A sequence starts at 1. Each step either doubles the previous term or adds the next positive integer, alternating operations. 1, 2, 4, 8, 12, 24, 31, ?. What is next?

- A. 38
- B. 42
- C. 48
- D. 62
- E. 64

Q38. Each term is $n^3 - n$ for $n = 2, 3, 4, \dots$ but one term is hidden by compression. Sequence: 6, 24, 60, 120, ?. What comes next?

- A. 180
- B. 210
- C. 240
- D. 252
- E. 336

Q39. The sequence alternates between adding a prime and multiplying by 2: +2, x2, +3, x2, +5, x2, ... Starting at 1: 1, 3, 6, 9, 18, 23, ?. What comes next?

- A. 28
- B. 36
- C. 40
- D. 46
- E. 51

Q40. A process tracks the number of possible states after n constrained moves. The state count sequence is 2, 3, 5, 8, 13, 21, ?. What is next?

- A. 29
- B. 31
- C. 32

D. 34

E. 35

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Batch 5: Q41–Q50 (Logic / Game Theory)

Q41. Exactly one of A, B, C is true. A says “B is false.” B says “C is true.” C says “A is false.” Who is true?

- A. A
- B. B
- C. C
- D. None
- E. Multiple

Q42. Three boxes are labeled Apples, Oranges, Mixed. All labels are wrong. Minimum number of fruit draws to determine all labels?

- A. 1
- B. 2
- C. 3
- D. 4
- E. Impossible

Q43. 100 prisoners and 100 boxes problem. Which strategy gives >30% survival?

- A. Random
- B. Pairing
- C. Cycle-following
- D. Split groups
- E. Recursion

Q44. Two doors, one truthful guard, one liar. Which question guarantees the safe door?

- A. Ask either directly
- B. Ask both separately
- C. Ask a nested/self-referential question
- D. Random guess
- E. Impossible

Q45. Hilbert’s hotel paradox demonstrates:

- A. Countable infinity
- B. Uncountable infinity
- C. Finiteness
- D. Logical error
- E. Physical impossibility

Q46. When is implication $P \rightarrow Q$ false?

- A. P false

- B. Q false
- C. P true and Q false
- D. Both true
- E. Both false

Q47. You have 9 identical balls, one heavier. Minimum weighings on a balance scale?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Q48. Two twins: one always lies, one alternates truth/lie each answer. Strategy to identify who is who?

- A. Single question
- B. Ask two questions exploiting alternation
- C. Impossible
- D. Random
- E. Symmetry trick only

Q49. Monty Hall with 3 doors. Best strategy after host opens a goat door?

- A. Switch
- B. Stay
- C. Same
- D. Depends on host
- E. Random

Q50. A set of statements is logically consistent if:

- A. No contradictions can be derived
- B. All statements are true
- C. Arguments are valid
- D. All are provable
- E. All are false

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Batch 6: Q51–Q60 (Coding / Algorithms)

Q51. You need to maintain the maximum of a sliding window of size k over an array in $O(n)$. Best data structure?

- A. Heap
- B. Deque
- C. Stack
- D. HashMap
- E. Segment Tree

Q52. You must maintain the median of a stream with insertions. Best approach?

- A. Sorting every time
- B. Two heaps (max + min)
- C. Balanced BST only
- D. Array + binary search
- E. DP

Q53. You are given a directed graph. Running topological sort fails. What can you conclude?

- A. Graph disconnected
- B. Graph has a cycle
- C. Graph has negative edge
- D. Graph not weighted
- E. Nothing

Q54. Which problem is best solved using Union-Find?

- A. Shortest path
- B. Connectivity queries
- C. Sorting
- D. Maximum flow
- E. DP optimization

Q55. Segment tree supports range query in?

- A. $O(n)$
- B. $O(\log n)$
- C. $O(1)$
- D. $O(n \log n)$
- E. $O(\sqrt{n})$

Q56. Dynamic programming is most suitable when:

- A. No overlapping subproblems
- B. Greedy always works

- C. Overlapping subproblems + optimal substructure
- D. Only recursion needed
- E. Randomized approach

Q57. BFS guarantees shortest path in:

- A. Weighted graph
- B. Unweighted graph
- C. DAG only
- D. Tree only
- E. None

Q58. Dijkstra's algorithm fails when:

- A. Graph is large
- B. Negative edge weights exist
- C. Graph is sparse
- D. Graph has cycles
- E. Source unknown

Q59. Extract-min in a binary heap takes:

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$
- E. $O(\sqrt{n})$

Q60. KMP algorithm is used for:

- A. Sorting arrays
- B. Substring search in linear time
- C. Graph traversal
- D. DP optimization
- E. Heap balancing